

Test of Mathematics for University Admission

Cambridge Assessment Admissions Testing

Simulation test

Paper 2 Time: 75 minutes

Additional Materials: Answer sheet

Instructions to Candidates

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. Please check you have one. You also require a soft pencil and an eraser.

This paper is the second of two papers.

There are 20 questions on this paper. For each question, choose the one answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

Any rough work should be done on this question paper. No extra paper is allowed.

Please complete the answer sheet with your candidate number, centre number, date of birth, and full name.

Calculators and dictionaries must **NOT** be used. There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This question paper consists of 21 printed pages and 3 blank pages.

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1 Two circles intersect at distinct points A and B . The tangents to the two circles at A are perpendicular. What is the angle between the tangents to the two circles at B ?

A 30°

B 45°

C 60°

D 90°

E It cannot be determined.

2 Let p be a real number. A sequence (a_n) is defined by an arbitrary real first term a_1 and

$$a_{n+1} = pa_n + (1 - p).$$

Which of the following conditions on p is/are individually sufficient to guarantee that (a_n) converges for every choice of a_1 ?

I. $|p| < 1.$

II. $p = 1.$

III. $p = -1.$

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

3 Let $p(x)$ be a polynomial. Consider the statement P :

$$p(x) = p(-x) \quad \text{for every real } x,$$

and the statement Q :

$$p'(x) = -p'(-x) \quad \text{for every real } x.$$

What type of condition is P for Q ?

A necessary and sufficient

B sufficient but not necessary

C necessary but not sufficient

D neither necessary nor sufficient

4 Starting with a positive real number t , repeatedly replace the current number u by $\log_2 u$ whenever $u > 1$. Stop as soon as the current number is at most 1.

Which condition is necessary and sufficient for the process to stop after exactly three replacements?

A $4 \leq t < 16$

B $4 < t \leq 16$

C $2 < t \leq 8$

D $8 < t \leq 16$

E $16 < t \leq 65536$

5 The non-zero real numbers a and b satisfy $\frac{1}{a} < \frac{1}{b}$. Which condition is necessary and sufficient for $a > b$?

A $a + b > 0$

B $ab > 0$

C $a^2 > b^2$

D $a > 0$

E $b < 0$

6 A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(f(x)) = x$ for every real x , and $f(1) = 3$. Which of the following functions must take the same value at two distinct real inputs?

A $f(x)$

B $f(x) - 2x$

C $f(x) - x$

D $f(x) + x$

E $f(x) + 2x$

7 For each real number t , let L_t be the line with equation

$$(t-1)x + (t+1)y = 2t.$$

For a fixed point (a, b) , which condition is necessary and sufficient for there to be exactly one real value of t for which (a, b) lies on L_t ?

A $a \neq b$

B $b \neq 1$

C $a + b \neq 2$

D $(a, b) \neq (1, 1)$

E $a + b = 2$ and $a \neq b$

8 Let x be a real number with $\cos x \neq 0$.

Which of the following conditions is equivalent to

$$\tan x > \sin x?$$

A $\sin x > 0$.

B $\cos x > 0$.

C $\sin x \cos x > 0$.

D $\sin x + \cos x > 0$.

E $\sin x \cos x < 0$.

9 Let a , b and c be positive real numbers. Consider the following statements.

- I. If a, b, c are consecutive terms of both an arithmetic progression and a geometric progression, then $a = b = c$.
- II. If a, b, c are consecutive terms of an arithmetic progression, then $1/a, 1/b, 1/c$ are consecutive terms of an arithmetic progression if and only if $a = b = c$.
- III. If a^2, b^2, c^2 are consecutive terms of an arithmetic progression, then a, b, c are consecutive terms of an arithmetic progression.

Which of these statements must be true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

10 Let $p(x)$ be a monic quadratic polynomial. Which of the following conditions is/are individually sufficient to guarantee that both roots of $p(x)$ lie strictly between 0 and 1?

I. $p(0) > 0$, $p(1) > 0$ and $p(1/2) < 0$.

II. $p(0) < 0$ and $p(1) < 0$.

III. $p(0)p(1) < 0$.

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

11 A student makes the following claim.

“If f and g both have a strict local minimum at $x = 0$, then $f - g$ has a local maximum or a local minimum at $x = 0$.”

Which of the following pairs of functions is a counterexample to this claim?

A $f(x) = x^2 + x^3, \quad g(x) = x^2$

B $f(x) = x^2, \quad g(x) = x^4$

C $f(x) = x^3, \quad g(x) = x^2$

D $f(x) = x^2, \quad g(x) = -x^2$

E $f(x) = x^4, \quad g(x) = x^2$

12 Let a, b be non-zero real numbers and let n be a positive integer.

The expansion of

$$(a + bx)^n$$

has exactly four negative coefficients.

Which of the following lists all possible values of n ?

A {3}

B {7, 8}

C {3, 6, 7}

D {3, 7, 8}

E {6, 7, 8}

13 A function $f : \mathbb{R} \rightarrow \mathbb{R}$ and real constants a, b satisfy

$$f(2x) = 3f(x) + x, \quad f(x+1) = f(x) + ax + b$$

for every real x . What is the value of $f(1)$?

A -2

B -1

C 0

D 1

E 2

14 Let $p(x)$ be a non-zero real polynomial and define $q(x) = p(x)^2$. Consider the following statements.

- I. At every simple real root of $p(x)$, the function q has a strict local minimum.
- II. At every repeated real root of $p(x)$, the function q has a strict local minimum.
- III. Every real number at which q has a local minimum is a root of $p(x)$.

Which of these statements must be true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

15 Let f and g be real-valued functions defined for all real numbers. A student claims that there is exactly one real number x for which $f(x) = 0$ and $g(x) > 0$.

Which of the following findings is/are individually sufficient to disprove the claim?

- I. The equation $f(x) = 0$ has no real solutions.
- II. The equation $f(x) = 0$ has at least two distinct real solutions.
- III. Whenever $f(x) = 0$, it follows that $g(x) \leq 0$.

A I only

B II only

C III only

D I and II only

E I and III only

F II and III only

G I, II and III

16 A continuous function f satisfies $f(x+2) = f(x)$ for all real x and $\int_0^2 f(x) dx = 0$. Define $F(x) = \int_0^x f(t) dt$. Which additional condition is sufficient to guarantee $F(x+1) = F(x)$ for every real x ?

A $f(x+1) = f(x)$ for every real x .

B $f(x+1) = -f(x)$ for every real x .

C $f(0) = 0$.

D $\int_0^1 f(x) dx = 0$.

E $f(2-x) = f(x)$ for every real x .

17 For each positive integer n , let $S_n = \sum_{k=1}^n a_k$. Consider the following statements.

- I. If S_n is a polynomial in n of degree 2, then (a_n) is an arithmetic sequence.
- II. If (a_n) is a non-constant geometric sequence, then (S_n) is a geometric sequence.
- III. If $S_{n+1} - 2S_n + S_{n-1}$ is the same non-zero constant for all $n \geq 2$, then (a_n) is a non-constant arithmetic sequence.

Which of these statements must be true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

18 Let p, q, r be primes such that

$$K = \frac{pqr}{p+q+r}$$

is an integer. Consider the following statements.

- I. K is prime.
- II. K is equal to at least one of p, q, r .
- III. At least one of p, q, r is at most 3.

Which of these statements must be true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

19 Let r be the positive solution of $r^3 + r = 1$. Choose t with $0 < t < r$. The tangent to $y = x^3 + x$ at $x = t$ meets the line $y = 1$ at a point with x -coordinate u . Which of the following must be true?

A $0 < u < t$

B $t < u < r$

C $u = r$

D $r < u < 1$

E $u \geq 1$

20 Let f be a monic quartic polynomial. The trapezium rule gives the same estimate for $\int_0^1 f(x) dx$ when either one strip or two equal strips are used. Let this common value be T , and let $I = \int_0^1 f(x) dx$. Which expression must equal $T - I$?

A 0

B $-\frac{1}{120}$

C $\frac{1}{120}$

D $\frac{1}{60}$

E It cannot be determined from the information given.